

■ SAFETY

ISOLATE AND LOCK OUT all electrical supplies before any maintenance. Minimum two persons for any work above 1.8 m. Never reach into the elevator casing while the belt/chain is in motion. Belt/chain may restart automatically if the VFD is in AUTO mode — confirm drive is in LOCAL/ISOLATED before opening any access door.

INSTALLATION SEQUENCE

Step	Activity	Details / Acceptance
1	Site preparation	Confirm foundation bolt pattern, level tolerance ≤ 2 mm over any 3 m. Confirm structural steel or concrete pedestal capacity per load schedule.
2	Boot section assembly	Install boot section and take-up. Level horizontally (< 0.5 mm/m). Fix boot anchor bolts — do not fully torque yet.
3	Casing erection	Erect casing sections bottom-up using craneage. Plumb to within 1:1000 (25 mm over 25 m height). Align all flanges, tighten bolts to 85 Nm (M16).
4	Head section installation	Lower head section onto top of casing. Align pulley centerline to boot centerline using laser or piano wire. Secure all flanges.
5	Drive installation	Mount motor and gearbox. Align coupling to ≤ 0.1 mm TIR. Fit coupling guard before any energisation.
6	Belt / chain installation	Thread belt/chain from boot to head. Install splice/attachment links. Adjust take-up to achieve initial tension (see commissioning procedure). Belt width 356 mm — ensure tracking margin on both sides.
7	Bucket installation	Attach buckets at the correct pitch. Verify correct orientation (open face up on carry side). Check attachment torque after first 4 hours of running.
8	Electrical connection	Connect motor and VFD per electrical schematic. Verify correct rotation BEFORE loading — jog in maintenance mode, confirm belt moves UP on carry side.
9	Final inspection	Check all access doors sealed. Confirm no tools/debris inside casing. Record pre-commissioning checklist below before energising under load.

PRE-COMMISSIONING CHECKLIST

#	Check	Accepted ✓	Sign / Date
1	All casing bolts torqued and checked		
2	Belt/chain splice/attachment links inspected		
3	Take-up tension set to design value		
4	Bucket attachment torque verified		
5	Drive alignment TIR < 0.1 mm confirmed		
6	Motor rotation confirmed (jog test — no load)		
7	VFD programmed: ramp-up ≥ 10 s, ramp-down ≥ 5 s		
8	All guards and covers fitted		
9	Feed chute clearance confirmed (no fouling)		
10	Emergency stop tested and documented		
11	Electrical isolation tagged out and released		

COMMISSIONING PROCEDURE

Step 1 — No-load run (15 min)

Start elevator empty. Run at 25 % speed for 5 min, then 50 % for 5 min, then 100 % for 5 min. Observe belt/chain tracking — adjust if edges contact casing. Record bearing temperatures: alarm > 70 °C, trip > 85 °C.

Step 2 — Partial load (30 min)

Introduce 50 % of design feed rate (139.7 t/h). Run for 30 min. Check for backlegging in boot, even bucket filling, and clean discharge at head. Record power draw — compare against design 37 kW.

Step 3 — Full load (60 min)

Increase to design capacity 279.3 t/h at design speed 1.83 m/s. Run for 60 min continuously. Record: power, bearing temps, belt tracking, any abnormal noise. Adjust take-up if tracking error increases.

Step 4 — Inspection after first load run

Stop elevator, lock out. Re-check bucket attachment torque. Inspect belt splice/chain links for wear. Check bearing lubrication. Re-torque any casing bolts that have loosened.

Step 5 — Sign-off

Complete all pre-commissioning checklist items. Obtain customer sign-off. Hand over this manual, maintenance schedule, and spares list.

MAINTENANCE SCHEDULE

Interval	Task	Category / Component
250h / 1wk	Belt tension re-adjustment after stretch (new belt only)	ADJUSTMENT
500h / 3wk	Belt visual inspection — splice, cover condition, edge fraying	INSPECTION
1,000h / 7wk	Belt tracking check and centralising adjustment	ADJUSTMENT
2,000h / 14wk	Check / top-up gearbox oil level	LUBRICATION
2,000h / 14wk	Take-up carriage travel and guide rail check	INSPECTION
2,000h / 14wk	Discharge chute clean-out and blockage check	CLEANING
2,000h / 14wk	Boot section clean-out (material spillage)	CLEANING
3,000h / 21wk	Re-grease head shaft bearings (3,000h interval)	LUBRICATION
3,000h / 21wk	Re-grease boot / tail shaft bearings	LUBRICATION
4,000h / 28wk	Lagging thickness measurement + adhesion check	INSPECTION
4,000h / 28wk	Casing panel and bolted joint inspection	INSPECTION
8,000h / 57wk	Gearbox oil change	LUBRICATION
8,000h / 57wk	Bucket condition + bolt torque check	INSPECTION
8,000h / 57wk	Drive coupling alignment check	INSPECTION

RECOMMENDED 1-YEAR SPARES LIST

Stocking these items on-site eliminates lead-time delays for the most common planned and unplanned replacement requirements. Quantities are based on the reliability model for this elevator configuration.

Component / Description	Est. Life (yr)	Action / Specification
Head and boot shaft bearings	17.3	Replace both bearing sets
Elevator belt (EP)	3.8	Full belt replacement + splice
Head pulley lagging (ceramic embedded rubber)	0.9	Strip and re-lag (hot vulcanised preferred) or replace pulley
Discharge chute wear liner	0.5	Replace liner panels — AR500 Wear Plate
Bucket set (full replacement)	5.8	Replace all buckets and attachment bolts simultaneously

Contact Jayvecons Engineering & Design for replacement parts, engineering support, or site service — reference the model number on the elevator nameplate which matches this document.